

Deep Learning-Based Approaches for Text Recognition in PCB Optical Inspection: A Survey

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Abstract—Automatic recognition of component markings and reference designators on a printed circuit board (PCB) is an important and challenging step for a high-accuracy automatic Bill of Materials(BoM) extraction, a hardware assurance technique. In this survey, we explore the performance of different state-of-the-art end-to-end text recognition approaches for automatic PCB marking recognition, along with the critical challenges associated with them. We also further emphasize the importance of collecting PCB marking data with appropriate annotations proposed in our earlier works. Moreover, we present some edge cases with essential insights for future work, along with open research capabilities and motivations for specialized end-to-end marking recognition algorithms for hardware assurance with self-explainability integration.

Index Terms—hardware assurance; bill of materials; optical character recognition; component markings and reference designators; deep learning; text recognition; self-explainable AI

I. INTRODUCTION

With growing advances in Internet of Things and Artificial Intelligence, PCBs have now become key components in many of these applications. Modern electronics, more so than before, now use mass-produced, mass-assembled components with micro electronic components outsourced from around the world [1]. With such a large supply chain, a natural consequence is the vulnerability of the hardware to attacks and counterfeiting. Over the years, counterfeit electronic components have been constantly increasing. In addition to various reports, The US Government has taken special interest due to the impacts of counterfeits on critical systems such as missiles, air crafts and defense systems [2] [3]. In addition, with the increase of defective and counterfeit components, industry [4] is also a victim and interested in new approaches to identify and authenticate the PCBs. With PCBs used for personal computing and communication purposes, privacy and data safety is vital for industries and their brand. In addition, they are critical in government, military, and biomedical applications, where reliability and security of the intellectual properties are a necessity. Thus, stringent safety measures are necessary and required to verify PCBs. PCB Component identification, authentication and assurance in general is vital problem that needs a better solutions.

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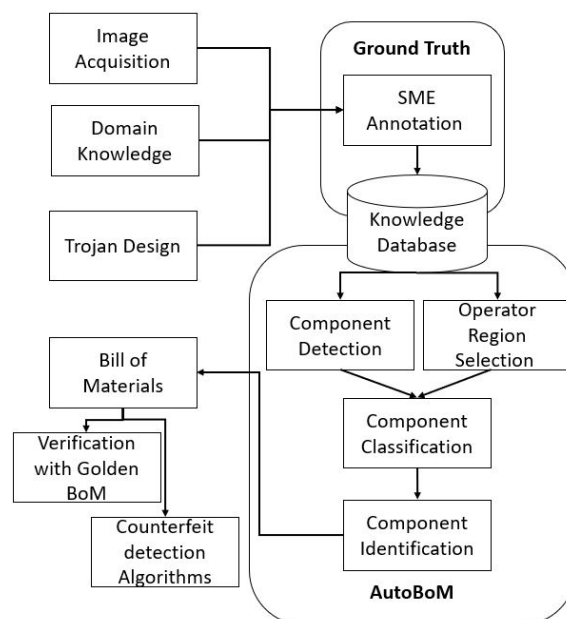


Fig. 1: The Automatic Bill of Materials Extraction Framework. With Domain knowledge and smart Image acquisition, the entire process of detecting, locating and identifying components, their serial numbers and data-sheets can be extracted and presented in a universal format such as BoM. This can be sent to further specialized algorithms of counterfeit detection and quality assurance

With the fast growth and developments in Artificial Intelligence(AI) and Parallel Computing, a novel framework for automatically extracting a PCB's BoM completely autonomously, was presented in 2019 [5]. This method proposes using a Bill of Materials (BoM) containing a list of components denoting Integrated Circuits, capacitors, and resistors, and other related details to these components. The extracted BoM could be used to compare with the Golden BoM(usually available from the design companies) for various scenarios like analysis of devices, security testing, design malware/defect detection and price estimation. This framework takes optical images of PCB's and markings such as Logos and text to extract the Bill of Materials. Fig. 1 presents an

illustration of the framework flowchart. An essential step in authenticating components, requires the identification of Text on the PCB. A significant number of recycled or counterfeit PCBs contain various inexpensive variants. These components are naturally, untested, unreliable or in very dangerous cases modified in functionality to aid an adversarial attack. Thus, in this manuscript we want to explore text detection in PCBs using existing Deep learning techniques. This will play a vital role in PCB assurance and authentication.

While we describe the various text detection methods in the related works section, we will briefly describe the dataset we use to re-train these popular network. The FICS-PCB dataset contains annotations for components, on-board text, and on-component text [6]. The PCB marking data collection on the FICS dataset comprises of about 9,912 images of 31 PCB samples and contains 77,347 annotated components.

In this study, our contributions are as follows: (i) We present a survey of existing text detection techniques, (ii) compare the two most popular techniques, with qualitative results, (iii) present challenges for existing text detection techniques that are difficult to solve, and motivate the need for specialized networks for the hardware security in general.

The manuscript is organized as follows. First, In Sec. II we discuss related works and relevant existing techniques. In Sec. III, we elaborate on the Hardware Security specific real-world challenges that need to be considered when addressing PCB Text detection. Following which, Sec. IV discussed about the specifics of the PCB text data set, including its collection, and augmentation setup. Sec. V presents quantitative results from testing popular approaches. Finally, We present a discussion on the challenges for the existing approaches and built on top of Sec. III in Sec. VI. Sec. VII concludes the study with summarising the manuscript's key takeaways and future directions that can improve the proposed work.

II. RELATED WORKS

Existing literature on Text detection can be classified into two categories. First category involves OCR techniques used for general purpose and scenery text recognition. The second category discusses existing approaches with applications in hardware security, especially for PCB level and IC level text detection.

A. General OCR

Conventional approaches for Text detection such as Tesseract [7], [8], MSER [9] have been introduced in the past. However, within the fast developing neural network approaches, text detection falls within the subset of the larger object detection problem. Here the difference being the object is a set of characters of the text. Using the popular idea of anchor-boxes introduced in [10], TextBoxes++ [11] presents an approach to identify multi oriented text. However this algorithm is sensitive to character spacing and thus misses out significant text. [12] presents a similar text detection approach using a U-net/ auto-encoder [13] with a Resnet [14] based architecture. Though robust to multi-orientations, it cannot handle curved text. Other

techniques such as [15] use sequence based learning with their proposed Sequence recognition network. The best performance in both detection as well as speed come from EAST [16], which performs text level detection with multi-orientations, and its proprietary improvement CRAFT [17]. We will discuss them in detail in under Sec. IV, under Methods.

In addition, Most of the networks are trained on popular datasets such as [11], [18], [19], however it can be noted that a majority of challenges for PCB text detection come from the dataset itself. A neural network is only as good as the data it is trained on. In addition, there can also be adversarial attacks on the system itself, which introduces further complexities, such as the one presented in [20]. There are numerous other approaches for Text detection, and they can be briefly identified in the literature described above, since most are compared with other similar approaches, but also in comprehensive survey papers at [21], [22]. On a related note, while different from text, logos detection is also of importance for PCB assurance. Works related to Logo detection can be found at [23]–[27]

B. PCB Specific OCR

While text detection in natural scenes and general Optical Character Recognition(OCR) are many, their application in PCBs is limited. [28] uses conventional approaches that use OCR engines including Tesseract [7] along with Stroke width techniques. [29] makes use of principal component analysis along with bayesian classification to achieve OCR. [30] presents a statistical analysis of different texts present in PCBs that create a fine background to understand PCB specific challenges for text detection.

Neural network based approaches to PCB text detection is even limited. This stems largely from the lack of data for PCBs and the lack of understanding the specific challenges in PCBs. [31] takes a step in this direction by introducing a dataset for PCB text detection. While they present initial results for text detection in PCBs, the results are far from perfect suggesting further research is required in this direction. [32] presents another similar approach for CNN based IC text detection, however with multiple false detection and scope for future work.

III. CHALLENGES FOR PCB SPECIFIC OCR

While text recognition in general is still an open problem with numerous challenges, PCBs pose further challenges unseen in the general problem. The challenges can be categorized into four based on the source of the challenge: (i) Challenges due to Acquisition Environment, (ii) Challenges due to the Components, (iii) Challenges due to the Board, (iv) Challenges due to the nature of PCB Text. Fig. 2 represents an exemplary case for each of these four challenges.

A. Challenges due to Acquisition Environment

Like every image based detection approach, PCBs also suffer from imaging challenges. However due to the high reflectivity in PCBs, the imaging setup is highly dependant

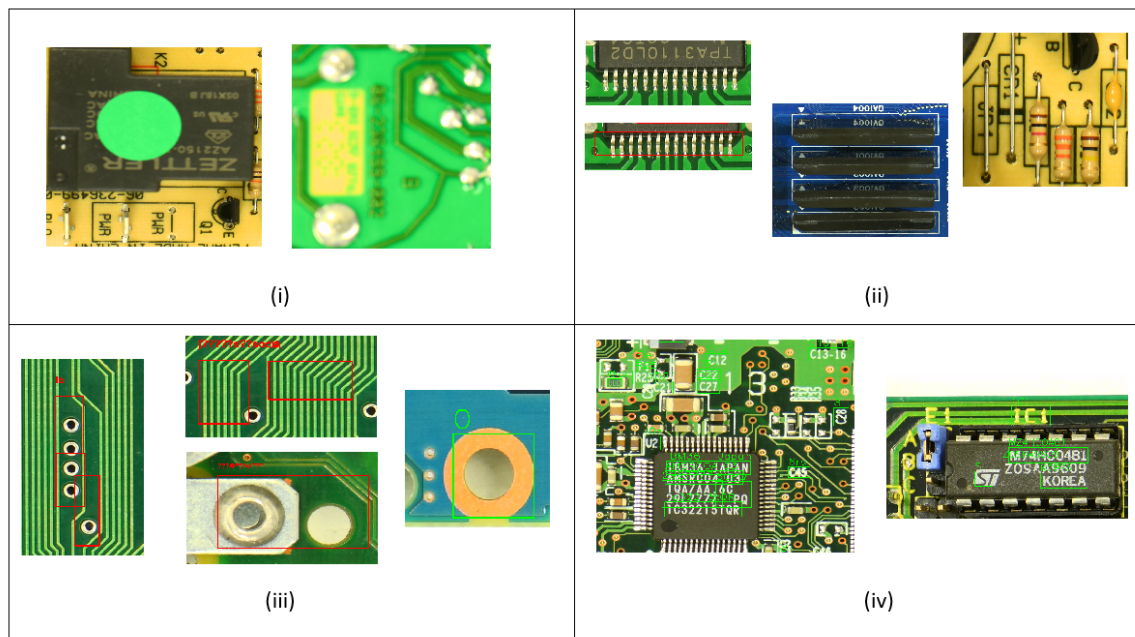


Fig. 2: Challenges for PCB Text detection

on saturation, exposure, focus and the contrast. Special care should be taken to modify these to fit the current imaging environment. In addition, due to the size of the components, the camera or imaging system should be able to image with a high resolution. The images in our dataset average around 4000 px x 3000 px in general to accommodate for the small components and text. Illumination challenges are one of the major challenges in PCBs, given the size of the PCB a uniform illumination should be maintained across the entire image. An example of this is shown in Fig. 2(i). This requires a dome setup, or a closed illumination setup which can be difficult to design and implement.

B. Challenges due to Components

In addition to the acquisition challenges, the components in the PCB board itself cause challenge. The most significant of these is occlusion due to tall components, leading to shadows occluding the text on the board nearby. Fig. 2(ii) shows an example of such an obstruction due to a shadow. Another challenge arises from through hole components which have a pin around them, that have a capacity to be categorized as a character ('1' or 'i' or 'l'). Though this is becoming less of a problem due to most modern PCBs having completely SMD components, it will definitely affect the algorithms application to legacy PCB boards. A final, but rare challenges comes from improper component assembly, where the component is placed on top of a reference designator. Though, the unforeseen problem could affect not only training a neural network, but also lead to false detection during testing.

C. Challenges due to Board

Since the PCB board can vary in color, this poses a challenge for text detection and neural network training. The board also includes various parts such as vias and traces, that also vary in color depending on the board substrate color. Vias are prime challenges since they are very often detected as characters, as can be guessed, as 'o' or '0'; Traces on the other hand are mostly detected as 'i' or 'l'. Overcoming this is very difficult, and would need a very carefully trained neural network. While there are certain markers around the vias such as a chrome yellow outline (similarly for a trace), these are very small to detect, and they occur only in certain colored PCB boards. Fig. 2(iii) presents examples of traces and vias that resemble characters. Finally, the texture of the board itself creates a problem. There are square or circular embedding in the Board, that are used for machine recognition of the board and its orientation during assembly. However, these cause a problem for text detection.

D. Challenges due to the nature of PCB Text

PCB text can be of two types - those on the board of the PCB called reference designators, and those on the surface of IC and other components, called component markings. Reference designators are useful to denote location of a component, whereas the component markings identify the component itself, making it possible to search the component in a database or over the internet. Both of these text are important, and each exhibit challenges for detection. The main problem with reference designators are its dense placement around the PCB. In a medium sized PCB (say, 5 x 5 inch PCB), there can be around 100 reference designators. This

number is significantly larger for larger PCBs. In addition, these also have a high chance to overlap with vias, traces further complicating the detection. Component level text on the other hand, pose a problem of varying font, and varying sizes. While the reference designators are all printed in one location, with the same font. The components are sourced from around the world showing different fonts and sizes. In addition, within a single IC, the font varies in sizes as well as font. Fig. 2(iv) shows an example of such a challenge. ICs in particular, also have Logos on them. While detecting Logos provide a way for additional authentication, detecting what is a Logo and what is Text challenge is difficult.

All of these challenges put together affect popular text detection approaches and their performance on PCBs. We will describe our text detection networks under investigation and their results in the following sections.

IV. METHODOLOGY OF EXISTING APPROACHES

A. Text Detection

Existing pieces of literature show that among all the regression-based approaches, EAST [16] performs the best considering detection, speed, and simplicity of the architecture. Despite being a complex and robust method, CRAFT [17] outperforms all the state-of-the-art techniques by predicting character regions with the affinity between the characters in a weakly supervised manner. That being the case, we have explored EAST [16] and CRAFT [17] as two of the most prospective approaches for detecting part markings and reference designators on full PCB images.

1) *Efficient and Accurate Scene Text Detector (EAST)*: This method comprises two simple stages. A Fully Convolutional Network (FCN) represents the first stage that produces words or text-line level predictions from the input image either in rotated rectangles (RBOX) or quadrangles (QUAD) as output. In the second stage, the predicted results are fed to the Non-Maximum Suppression to yield the final results. The structure of the FCN can be divided into three parts: a feature extractor stem which can be PVANet [33] or VGG16 [34] pre-trained on ImageNet [35] dataset, a feature-merging branch with four levels of feature maps, and an output layer fed by the final feature map which is a CONV 3x3 layer [16].

2) *Character Region Awareness for Text Detection (CRAFT)*: The overall idea of CRAFT includes localizing the individual character regions and linking them to a text instance by predicting two scores: region score, which localizes the character, and affinity score, which merges characters into a single text instance or word. This method adopts a fully convolutional network (FCN) architecture based on VGG16 [34] with batch normalization as its backbone. The decoding portion of the CRAFT architecture is similar to the U-Net [36] architecture. It consists of skip connections that aggregate low-level features [17].

B. Text Recognition

1) *Convolutional Recurrent Neural Network (CRNN)*: CRNN [37] is one of the few common neural network-based

architecture used in OCR. It is a combination of Deep Convolutional Neural Networks (DCNN) and Recurrent Neural Network (RNN). The model comprises three components: the convolutional layers, the recurrent layers, and a transcription layer. The purpose of the convolutional layers is to extract a feature sequence automatically from each input image. A recurrent network is built on top of the convolutional network for making a prediction for each frame of the feature sequence generated by the convolutional layers. The transcription layer translates the per-frame predictions by the recurrent layers into a label sequence. CRNN can be jointly trained with one loss function such as CTC (Connectionist Temporal Classification) loss.

2) *Tesseract OCR Engine*: Another deep-learning-based text recognition approach with both a high accuracy level and exposure to a wide variety of languages is Tesseract OCR Engine [8] from Google. The first step of Tesseract OCR represents a connected component analysis where the outlines of the components are stored by nesting into blobs. Blobs are then organized into text lines, and the lines and regions are analyzed for fixed pitch or proportional text. Text lines are divided into words according to the types of character spacing. After that, the recognition stage involves a two-pass process. In the first pass, each word is attempted to be recognized in turn. An adaptive classifier takes each satisfactory word as training data. In the second pass, an attempt is made again to recognize previously unrecognized words. Finally, fuzzy spaces are resolved, and small-cap texts are located using alternative hypotheses.

C. Evaluation Metrics

There is more than one evaluation metric for each of three OCR-related tasks: detection, recognition, and end-to-end system. However, no single metric can address all of the OCR challenges efficiently [38]. Since there is no single evaluation metric which works for all the existing approaches and the detailed analysis of the metrics is beyond the scope of this paper, we have confined our result analysis to measuring precision, recall, and F-measure, which are defined as below.

$$Precision = \frac{\text{No. of Correctly Recognized Texts}}{\text{Total No. of Recognized Texts}} \quad (1)$$

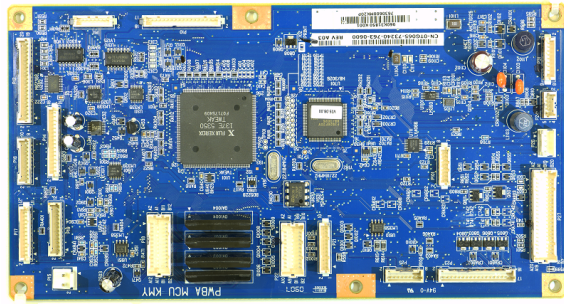
$$Recall = \frac{\text{No. of Correctly Recognized Texts}}{\text{Total No. of Ground Truth Texts}} \quad (2)$$

$$F - \text{measure} = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (3)$$

D. Modifications: Windowing Approach

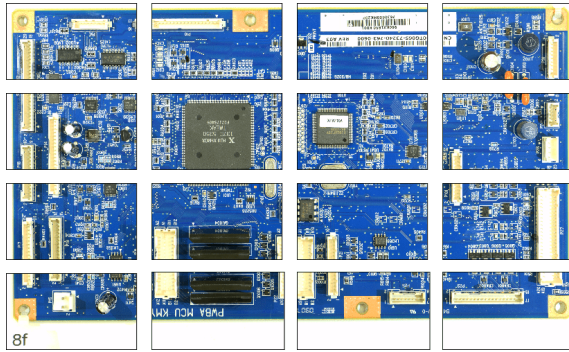
In the context of computer vision, one popular approach in object detection is using sliding window algorithm. We have adopted this idea but modified the technique according to our problem. As discussed in Sec. III, due to the variable density of reference designators in full PCB image, it is quite

challenging for an OCR model to recognize all the reference designators. It holds for some cases of component markings as well. Since state-of-the-art end-to-end methods have done quite well in detecting and recognizing texts in natural scene images where the density of text regions are lower compared to full PCB images, we divided a full PCB board image into an equal number of blocks/windows of the same size and then implemented the detection and recognition technique and finally combined the separate blocks with recognized component markings and reference designators to reconstruct the full PCB image. In this work, we set the number of blocks



8f

(a) Sample full PCB image.



(b) Dividing the full PCB image into 16 equal-sized blocks.

Fig. 3: Modified windowing approach for full PCB OCR

to 2^n , where $n = 1, 2, 3, 4$. The number of blocks was limited to 16 ($n = 4$) since an increasing number of blocks created overlapping of detected text regions and often broke one text instance into several parts, resulting in ambiguous recognition results. Moreover, it takes longer inference time when the number of blocks increases. A sample image divided into 16 equal-sized blocks is shown in Fig. 3(b).

V. EXPERIMENT

A. Datasets

As explained in Sec. III, there are different key challenges for automatic PCB marking detection and recognition, the majority of which can be addressed by using a large and adequately labeled dataset. However, the lack of available large PCB marking datasets in the hardware assurance community has made the solutions currently unattainable. Though there are many benchmark datasets for logo and text detection and

recognition in other domains (e.g., text in natural scenes [39], logos of commercial brands [40] and clothing [41], etc.), few of them represent some of the challenges in the PCB domain. Therefore, a large PCB marking data is being collected in the capacity of the Florida Institute for Cybersecurity (FICS) Research at the SeCurity and AssuraNce (SCAN) Lab at the University of Florida as a part of an ongoing multi-modality PCB data collection effort [6]. We have used 34 full high-resolution PCB images with a total of 6089 component markings and reference designators for our experiment. The highest no. of component markings and reference designators (696) are present in image id 8, and the lowest no. of component markings and reference designators (1) are present in image ids 20 and 25. Among the 34 full PCB images, there are 25 green boards, 4 blue boards, 2 red boards, 2 black boards, and 1 yellow board. The overview of the dataset is illustrated in Fig. 4. All images are in .png format and average resolution is around 4000 px x 3000 px. The annotations are in ICDAR [19] format.

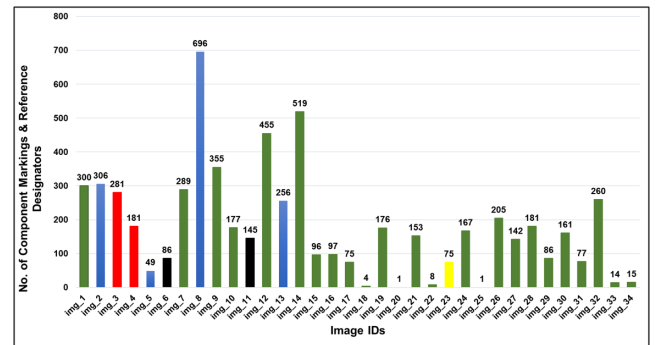


Fig. 4: Brief overview of the full PCB Image Dataset for our experiment

B. Implemented Pre-trained Models

Two end-to-end frameworks were implemented on the dataset. First framework is called EasyOCR developed by JaidedAI [42], which incorporates pre-trained CRAFT algorithm [17] for detection part and CRNN [37] for recognition part. On the other hand, the second framework uses pre-trained EAST algorithm [16] for detection part and tesseract OCR engine [7] for recognition part.

C. Experimental Results

Both the frameworks were implemented without and with windowing approaches. The produced results are analyzed in the following subsections.

1) *End-to-End Framework-1*: A decreasing trend in precision was observed with the increase of the number of blocks. In contrast, an increasing trend was observed in recall with the increase in the number of blocks. F-measure was the highest for no. of blocks = 16. But there was a time-performance trade-off. The higher the number of blocks, the longer the inference time. Table I shows the overall results for End-to-End Framework-1.

2) *End-to-End Framework-2*: For End-to-End Framework - 2, the windowing approach did not help improving the performance as reflected in Table I. A downward trend was observed for all the performance measures with the increasing number of blocks. The worst performance was observed for the number of blocks = 8. Moreover, the windowing approach required a longer inference time.

3) *Comparison between the frameworks*: Table I shows a comparative analysis of both the end-to-end frameworks implemented on our PCB image dataset. It is evident from the analysis that though the number of recognized component markings and reference designators increases by applying the windowing approach, the number of correctly recognized component markings and reference designators is far too low compared to it. The situation is worse in the case of the second end-to-end framework. It was expected because none of the detection algorithms were trained with PCB images. Since the characteristics of PCB image data is far different from natural scene images, and there exist lots of challenging scenarios in PCB marking data compared to scene-text data, it can be expected that training the existing algorithms or a new algorithm with a large PCB image dataset with appropriate annotations considering the challenging scenarios will enhance the marking detection and recognition performance.

Our focus on this manuscript extends to only a comparison of the performance of these frameworks. We cannot provide a computational analysis and comparison because these networks were trained on different datasets, under different imaging conditions, varying hardware capabilities and with differing perspectives.

VI. DISCUSSION

A. Edge Cases

After conducting a qualitative analysis of the produced results, some edge cases were found which should be considered while developing an end-to-end framework for PCB marking detection and recognition. Both the frameworks faced difficulties while detecting and recognizing marking in the following scenarios: (i) Same board, but different orientation of component markings and reference designators (as shown in Fig. 5(i)), (ii) Vias, pins, traces, solder pads/joints, tags, and sticker markings recognized as component markings and ref. designators (as shown in Fig. 5(ii)), (iii) Variable font style and size (as shown in Fig. 10(iii)), (iv) Density of component markings and reference designators (as shown in Fig. 5(iv)), (v) Reference designators under the components (as shown in Fig. 5(v)).

B. Insights

From qualitative and quantitative analysis, some key insights can be obtained as following: (i) The windowing approach has two drawbacks despite improving recognition performance: first, it is time-consuming; second, it sometimes may break one word into multiple portions, and the end-to-end framework may not detect and recognize them as a single word, (ii) The impact of board color can not be clearly understood from the

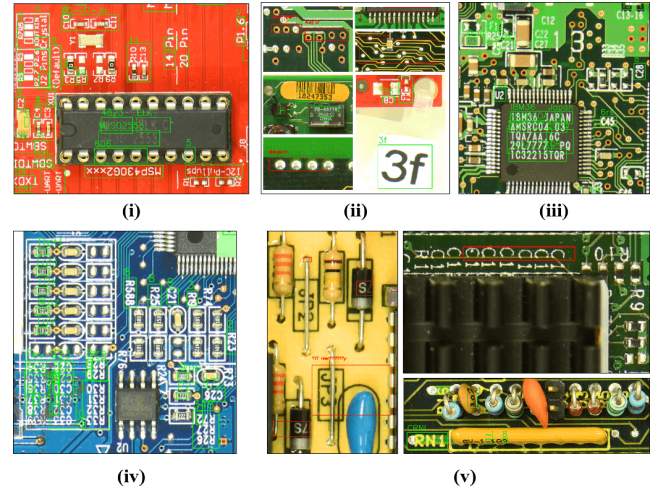


Fig. 5: Edge cases for PCB marking detection and recognition

performance of both the frameworks, (iii) Presence of excessive vias and traces impacted the performance of end-to-end framework-2, but it is not the case for end-to-end framework-1, (iv) The higher the density of component markings and reference designators, the higher the number of incorrectly recognized component markings and reference designators, (v) Character-region level awareness (End-to-End Framework-1) is more suitable for PCB OCR than word-level detection (End-to-End Framework-2) because the former one is robust but more accurate. In contrast, the latter one is faster but less accurate, (vi) For both the frameworks, the best performance was observed for image_id = 29. Analyzing the characteristics of the board, it is found that the corresponding PCB board has no component markings, all are board markings with a uniform background (green board) which is illustrated in Fig. 6. Therefore, a mixture of component markings and reference designators with non-uniform background is a challenging case for PCB OCR.

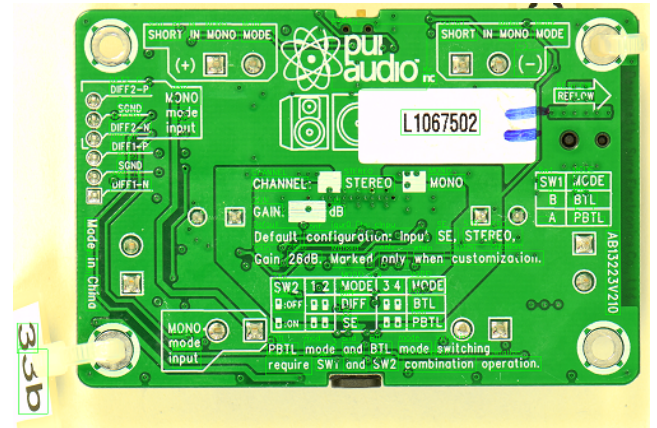


Fig. 6: PCB Image without component markings yields better performance (sample result for Image ID: 29 using End-to-End Framework-1).

TABLE I: Comparative Analysis of Both End-to-End Frameworks Implemented on the PCB Image Dataset

Frameworks		End-to-End Framework-1						End-to-End Framework-2					
Approaches		Total GT	Total Recognized	Total Correctly Recognized	Precision	Recall	F-Measure	Total GT	Total Recognized	Total Correctly Recognized	Precision	Recall	F-Measure
Without Windowing		6089	3462	220	0.063547	0.036131	0.046068	6089	1455	56	0.038488	0.009197	0.014846
With Windowing	No. of blocks = 2		3723	220	0.059092	0.036131	0.044843		1436	26	0.018106	0.00427	0.00691
	No. of blocks = 4		4054	233	0.057474	0.038266	0.045943		2215	44	0.019865	0.007226	0.010597
	No. of blocks = 8		4070	231	0.056757	0.037937	0.045477		2745	21	0.00765	0.003449	0.004754
	No. of blocks = 16		4356	243	0.055785	0.039908	0.046529		3368	47	0.013955	0.007719	0.00994

C. Self-Explainable AI for PCB OCR

An efficient automated hardware assurance (e.g., Automated PCB assurance, AutoBoM) should not require any SME (Subject Matter Expert) while performing a task like PCB OCR. Therefore, any developed stand-alone framework for PCB OCR should provide confidence levels for both their output and their explanation which is why the exploration of a self-explainable deep learning model for PCB OCR is important. Moreover, according to Daniel C. Elton (Author of [43]), “applicability and domain analysis is especially important for AI systems where robustness and trust are important so that systems can alert their user if they are asked work outside their domain of applicability.” Therefore, to develop AI systems for hardware assurance that can be trusted, the integration of self-explainability is of utmost importance.

D. Future Research Directions

Based on the above discussions, following future research directions can be implied:

- Development of a large PCB image dataset
- Development of a better end-to-end framework for PCB OCR offering solutions to the existing challenging cases
- Enabling the algorithm to explain the reasoning behind its performance and decision without the presence of SME (integrating self-explainable AI)
- Establishing a suitable metric to efficiently evaluate the performance of the algorithm
- Addressing the speed-accuracy trade-off of the developed stand-alone framework

VII. CONCLUSIONS

This survey presents an implication that the hardware assurance community should work on the development of a specialized end-to-end marking recognition algorithm for recognizing component markings and reference designators on PCBs. In addition, we also discuss the importance of a large amount of PCB marking data with corresponding ground truth data. In this study, we elaborated on the existing challenges for PCB marking recognition. This is also reflected in the edge cases subsection under Sec. VI. We obtain and present some important insights after conducting our experiments using existing approaches. These insights will motivate our future efforts in developing a large PCB marking dataset for a high-accuracy automatic Bill of Materials (AutoBoM) extraction.

Our future endeavors focus on include collecting and annotating PCB marking data, developing an end-to-end marking recognition algorithm addressing the edge cases, with the capability of incorporating self-explainability as discussed in

Sec. VI. These resources and efforts will help the community to achieve significant progress towards materializing automated hardware assurance.

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